



Optimization Tools

V.0.1.1 manual
(Chinese version)



什么是Optimization tools

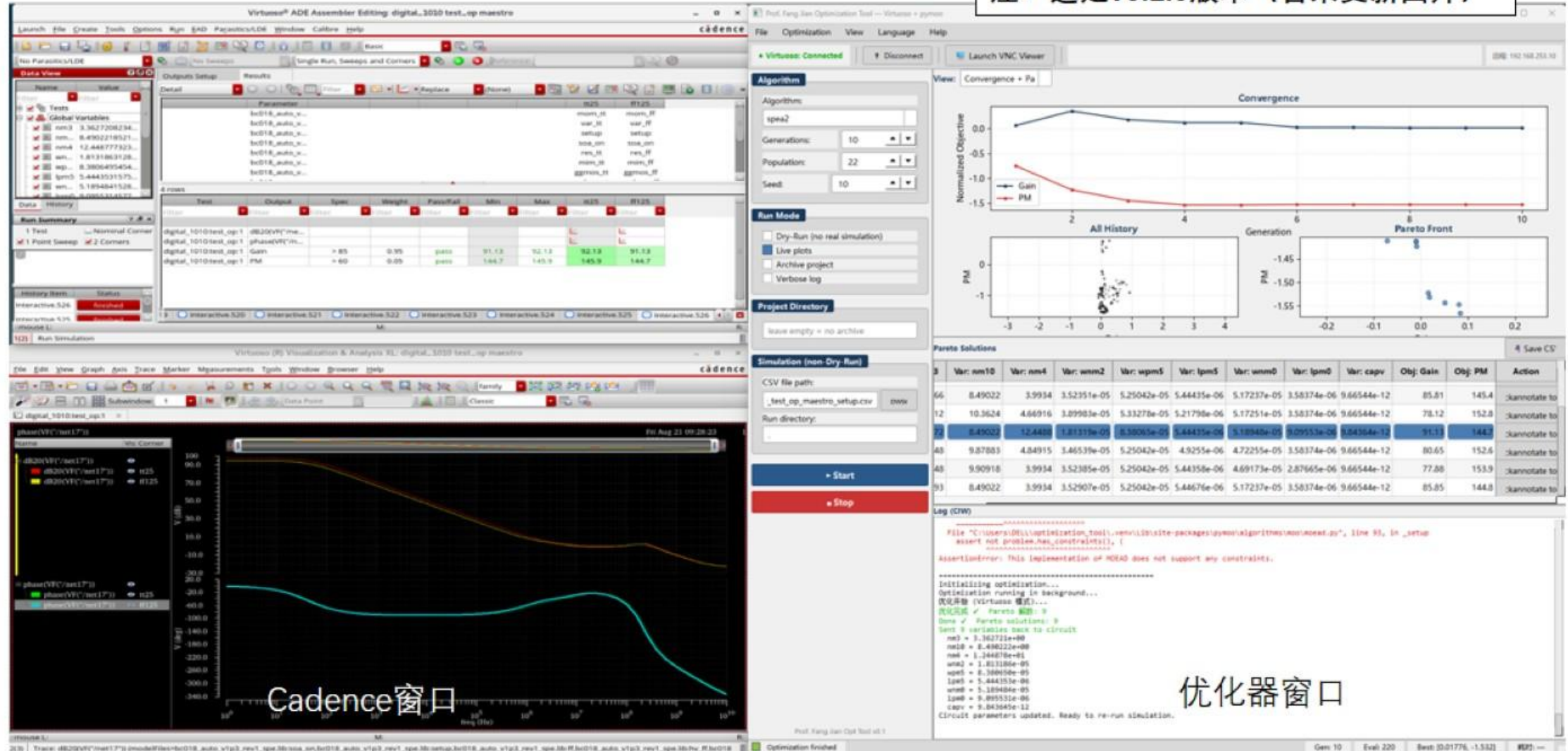
Optimization tools 是自主开发的、一个外挂于cadence的电路参数优化器。可以本地或远程接入运行中的cadence，帮助cadence完成电路参数优化。与有以下潜在的优势：

- 优化算法更丰富；
- 便于接入AI实现下一代参数优化、拓扑优化；
- 具有开放性，可迅速接入新算法；
- 提供分布式并行计算的能力，不完全依赖cadence自身的APS机制。



Optimization tools界面:

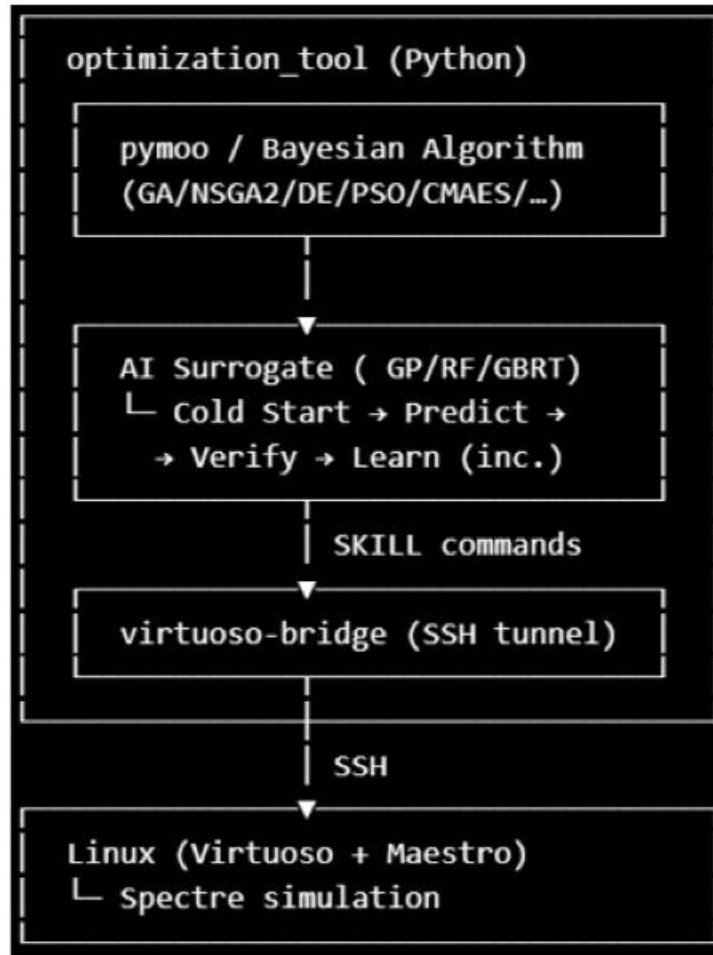
注: 这是v0.1.0版本 (暂未更新图片)





Optimization tools主要算法支持:

支持多参数, 多目标电路调参优化



Available Algorithms

Category	Algorithms
Single-objective	ga, de, pso, cmaes
Multi-objective (2-3)	nsga2, spea2, nsga3, moead, rnsga3
Many-objective (4+)	nsga3, moead, ctea
Bayesian (single)	bayes_gp, bayes_rf
Bayesian (multi)	bayes_mo

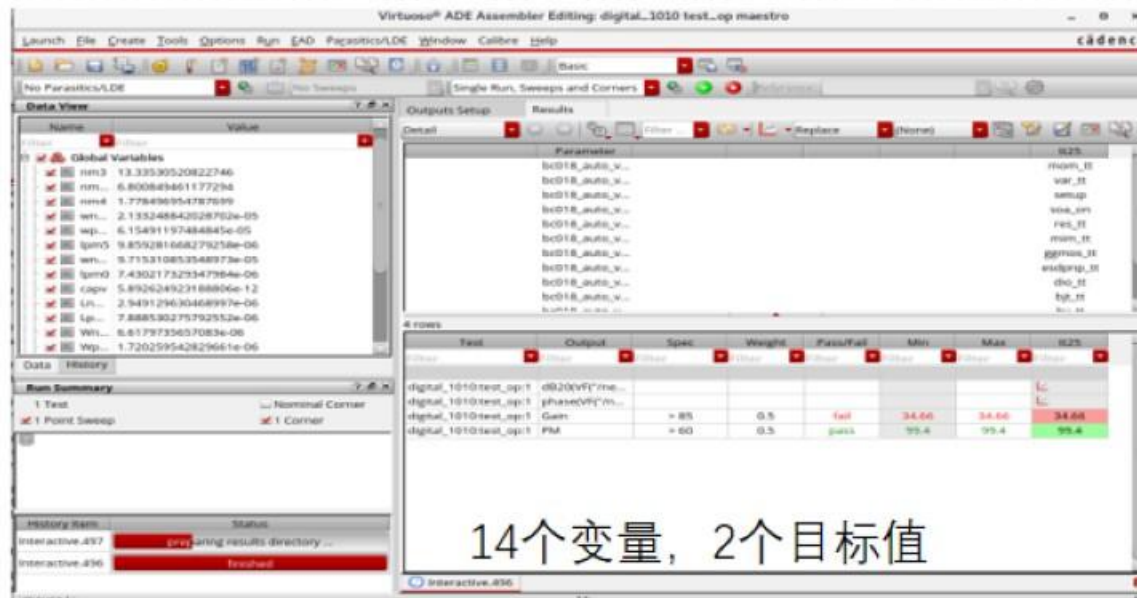
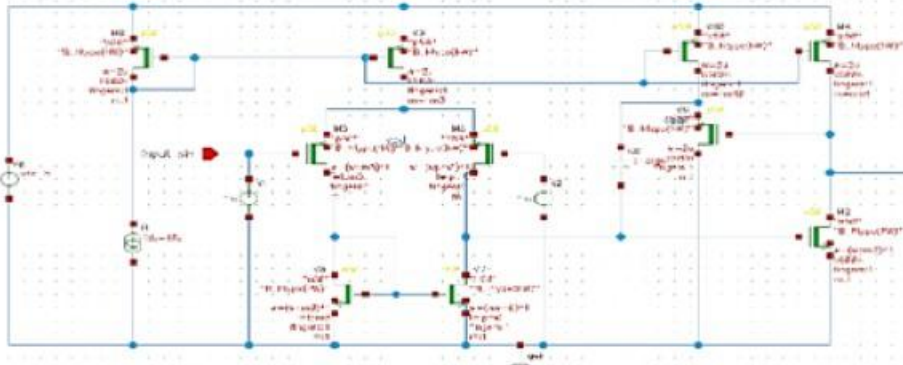
AI Surrogate Assistance

The AI surrogate module accelerates optimization by replacing costly Spectre simulations with fast model predictions. It supports three phases:

Model	Type	Best For
local_gp	Gaussian Process (RBF/Matern kernel)	Smooth response surfaces, small samples
random_forest	Random Forest regression	Robust to outliers, medium samples
gradient_boosting	Gradient Boosting + quantile	Non-smooth responses, large samples



运行实例（两级OTA OP）：



14个变量，2个目标值

3.3. GP+NSGA2 (继承+冷启动样本8)

[AI 统计]

总计评估: 80

真实仿真: 8

虚拟预测: 72

AI 代理率: 90.0%



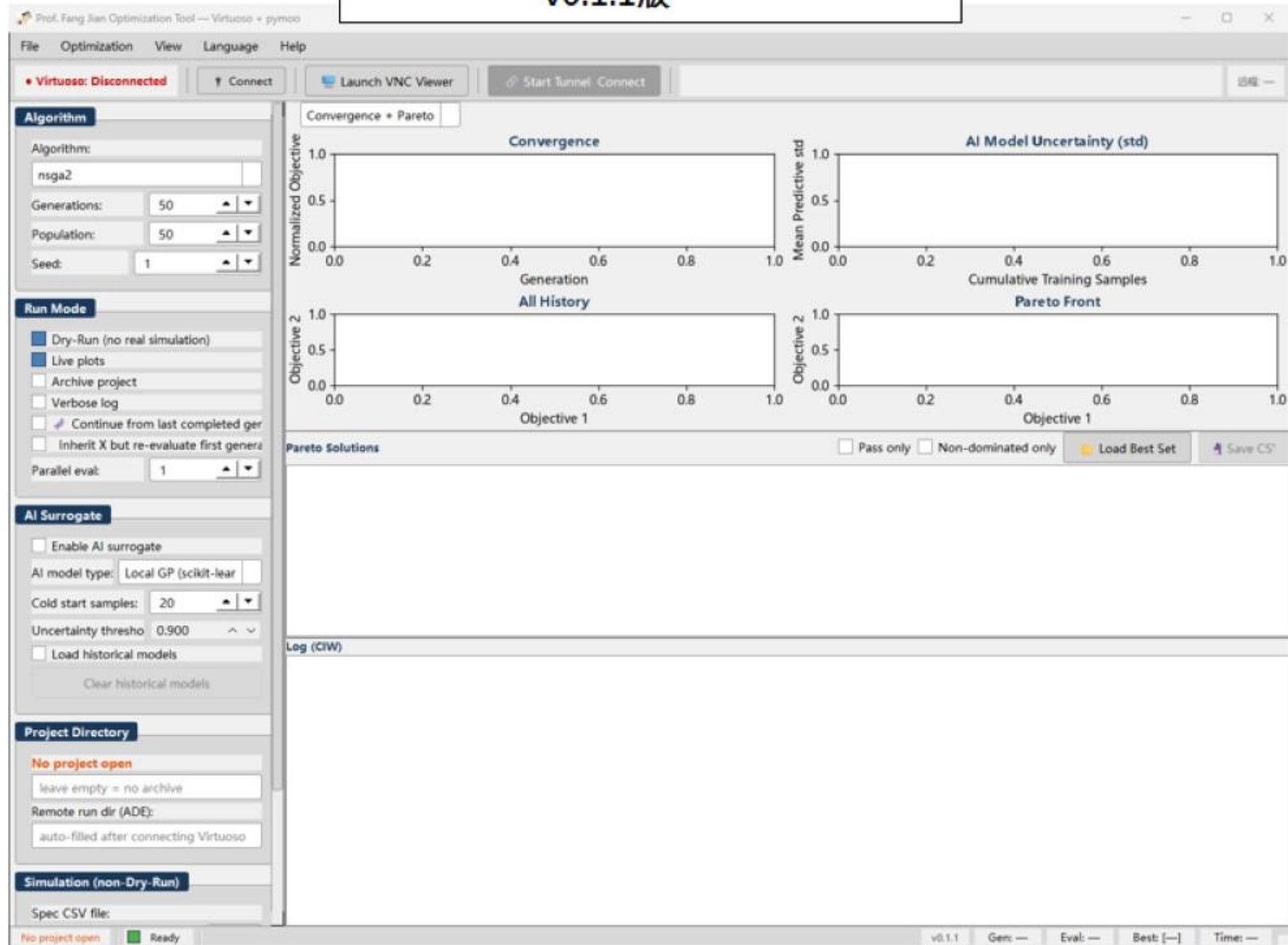
最优解集中有满足spec的数据。



现有版本的使用主要步骤



v0.1.1版





使用主要过程

1. 建立SSH隧道
2. Cadence侧完成优化设置
3. 运行Optimization
4. 完成优化后参数的回调



②连接Virtuoso工作窗口

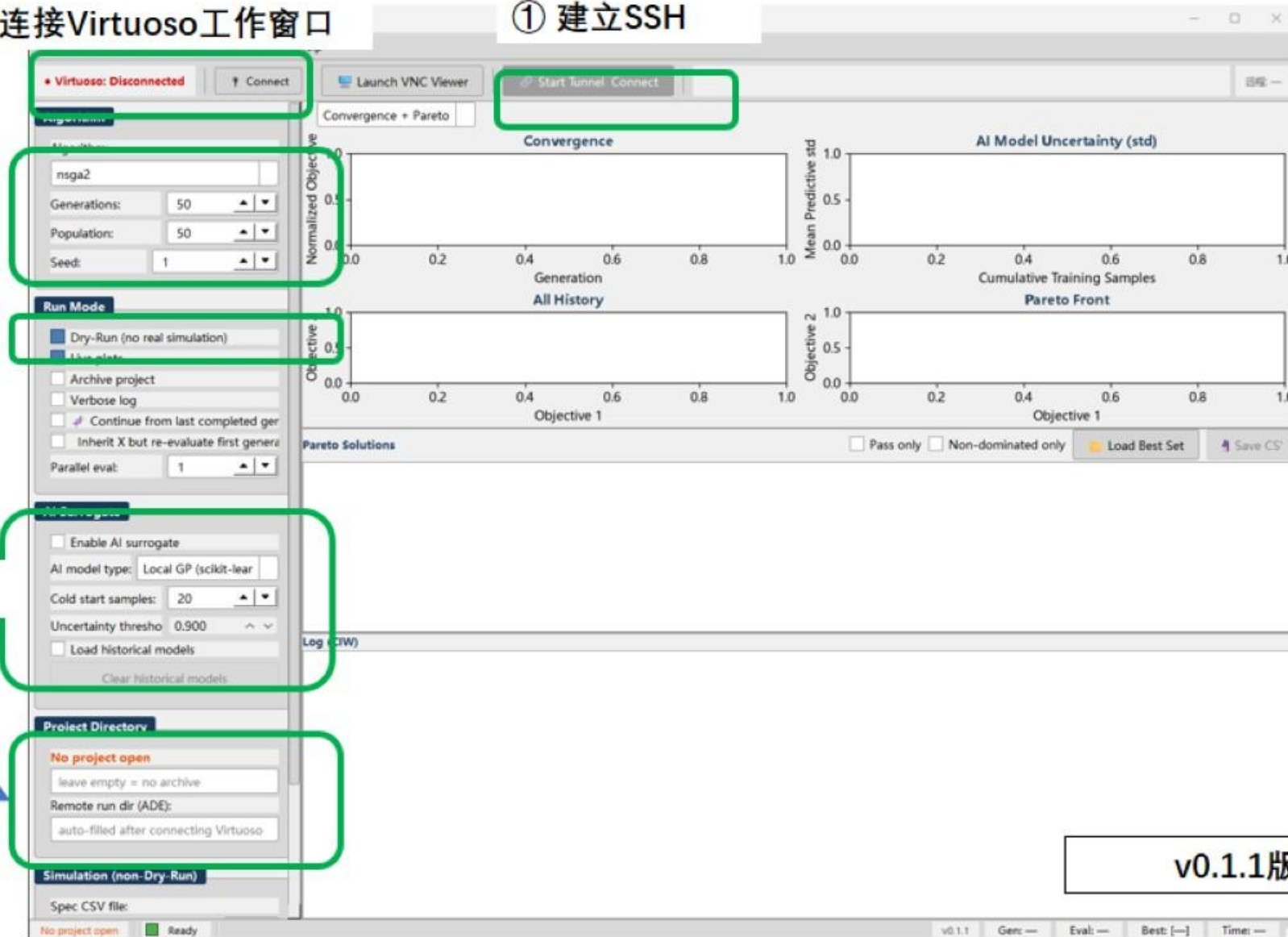
① 建立SSH

③设置算法

④关闭Dry；运行实际电路

③设置算法

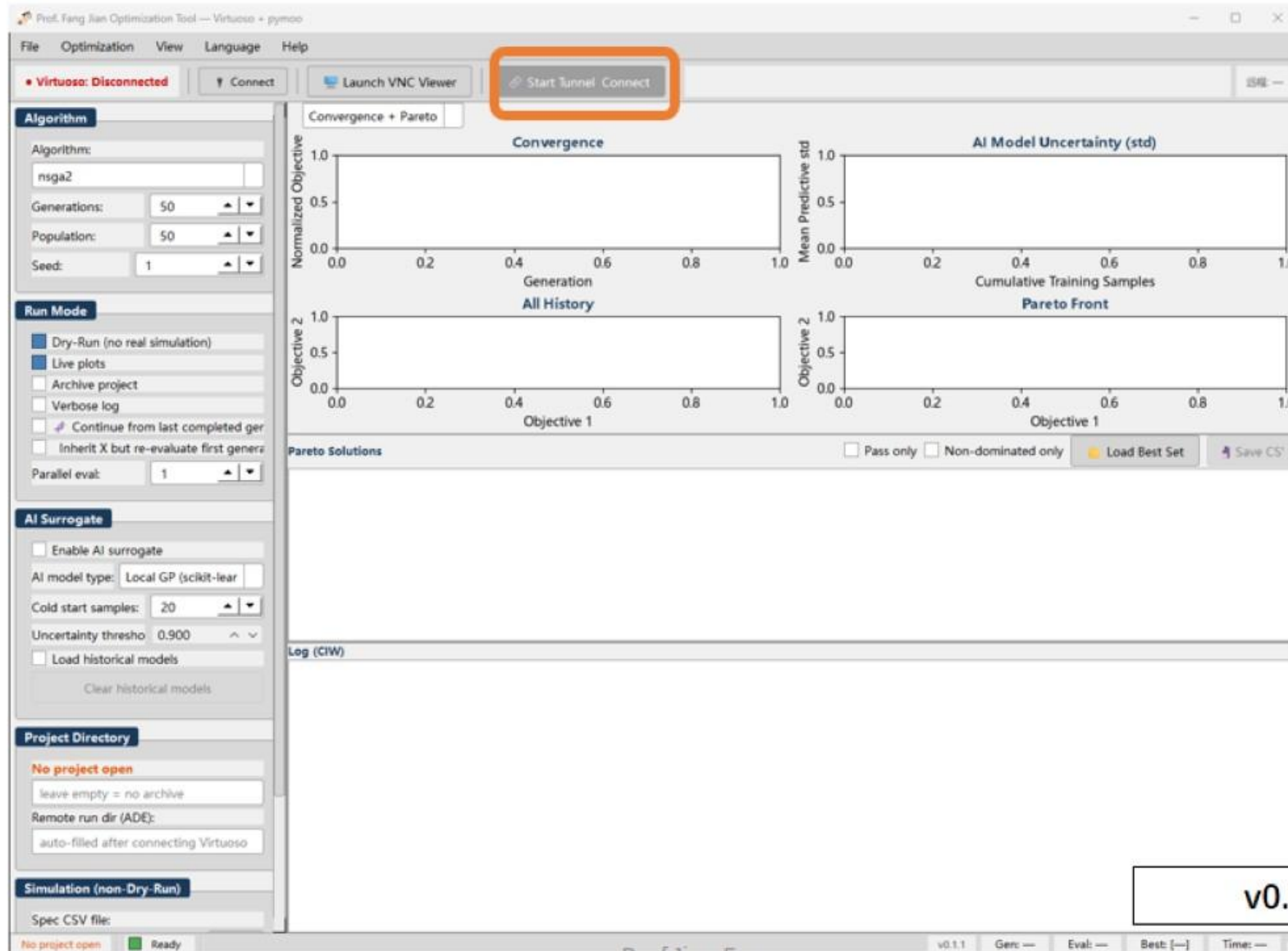
⑤加载目标 (spec) 文件 (目前需手动copy到本地)



v0.1.1版



2. 建立SSH隧道



v0.1.1版



+ FullyQualifiedErrorId : UnexpectedToken

```
PS C:\Users\DELL\virtuoso-bridge-lite-main> .\start_bridge.bat
```

```
=====
Virtuoso Bridge Launcher
=====
```

```
Enter remote host (default 192.168.1.100)
```

```
Enter remote user (default root)
```

```
=====
Please run this command in Virtuoso CIW:
```

```
load("/tmp/virtuoso bridge colon/DELL/virtuoso bridge/virtuoso_setup.il")
```

```
Note: Skip if already loaded in CIW
```

```
-----
Virtual environment is active. You can run Python commands now.
```

```
Type "exit" to close this window.
```

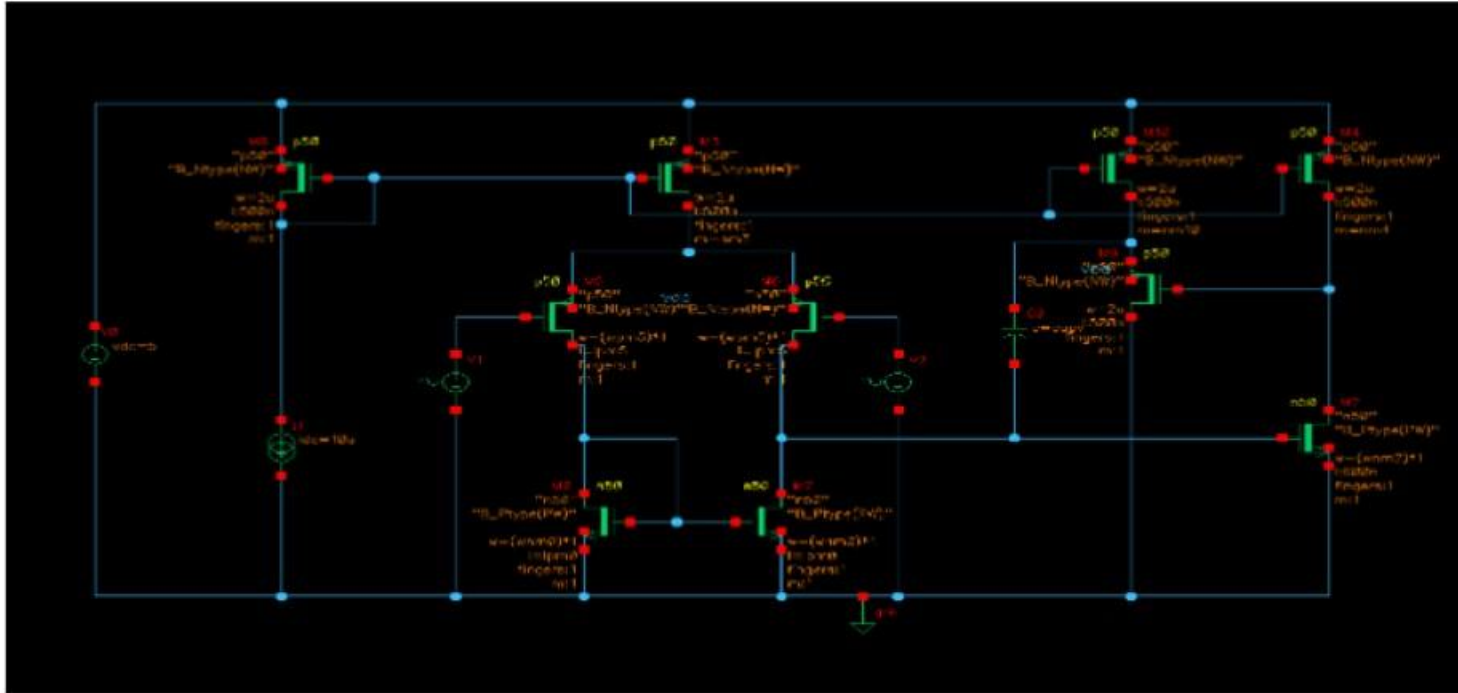
```
===== All the best! =====
```

复制这段代码到cadence CIW中执行

这样SSH隧道就建立好了



首次建立SSH隧道，需要合法的用户及密码（linux）。



绘制电路图，图中需要优化器件的参数需要用变量给出。例如：

Edit Object Properties

Apply To: ☒ only current ☒ instance

Show: ☐ system ☒ user ☒ CDF

Browse Reset Instance Labels Display

Property	Value	Display
Library Name	smic18bcdp1	off
Cell Name	p50	value
View Name	symbol	off
Instance Name	M5	off

Add Delete Modify

CDF Parameter	Value	Display
Model Name	p50	off
Multiplier	1	off
Length	1pe5 M	off
Total Width	1Par("fw") * 1Par("fingers") M	off
Finger Width	1pe5 M	off
Fingers	1	off
Threshold	300n M	off
Apply Finger Width Threshold	☐	off
Gate Connection	☒ None ☐ Top ☐ Bottom	off
Gate Direction	☒ Mid ☐ Left ☐ Right	off
S/D Metal Width	230n M	off
Switch S/D	☐	off
Bodytie Type	None	off
Edit Area & Region	☐	off



(3.1) 打开ADE Assemble仿真环境

The screenshot shows the Cadence Virtuoso ADE Assembler Editing window for a project named 'digital_1010 test_op maestro'. The interface includes a menu bar, a toolbar, and several panels. Annotations 1 through 4 highlight specific areas:

- ① Points to the 'Parameters' section in the left-hand 'Data View' panel.
- ② Points to the 'Global Variables' section in the left-hand 'Data View' panel.
- ③ Points to the 'Outputs Setup' table in the right-hand panel.
- ④ Points to the 'Single Run, Sweeps and Corners' button in the top toolbar.

Outputs Setup Table:

Filter	Variable	Type	Expr	Unit	Filter	Value	Filter	Value	Filter	Value	Filter	Value
digital_1010:test_op:1		expr	dB20(VF("/net17"))	point								
digital_1010:test_op:1		expr	phase(VF("/net17"))	point								
digital_1010:test_op:1	PM	expr	phaseMargin(VF("/net17"))	point								
digital_1010:test_op:1	Gain	expr	value(dB20(VF("/net17")) 1)									

Run Summary:

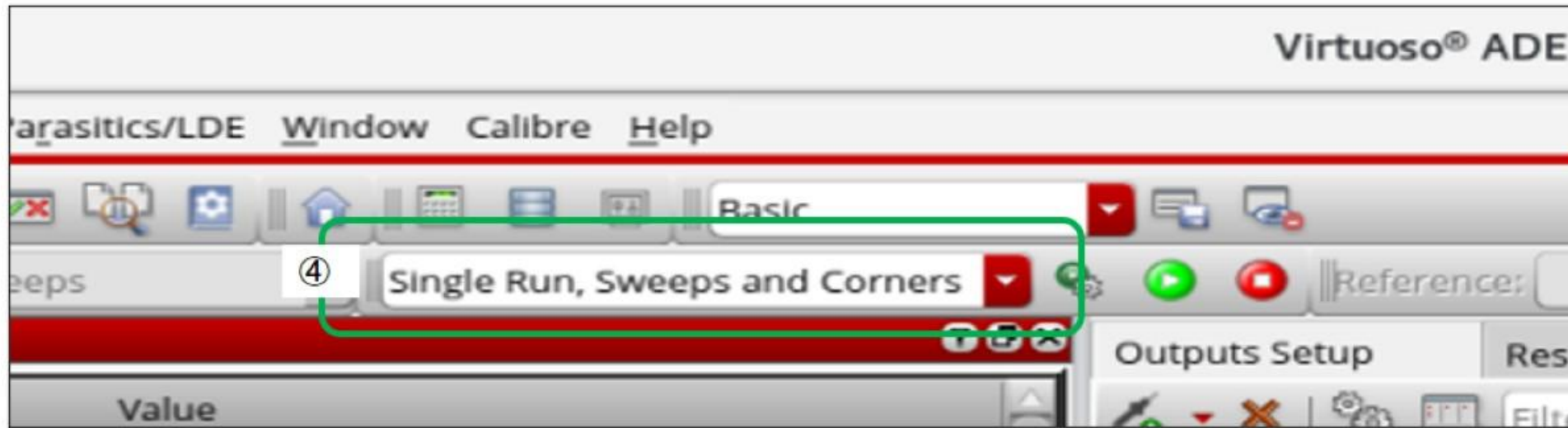
- 1 Test
- 1 Point Sweep
- 2 Corners

History Item:

History Item	Status
Interactive:186	finished
Interactive:185	finished
Interactive:184	finished
Interactive:183	finished
Interactive:182	finished

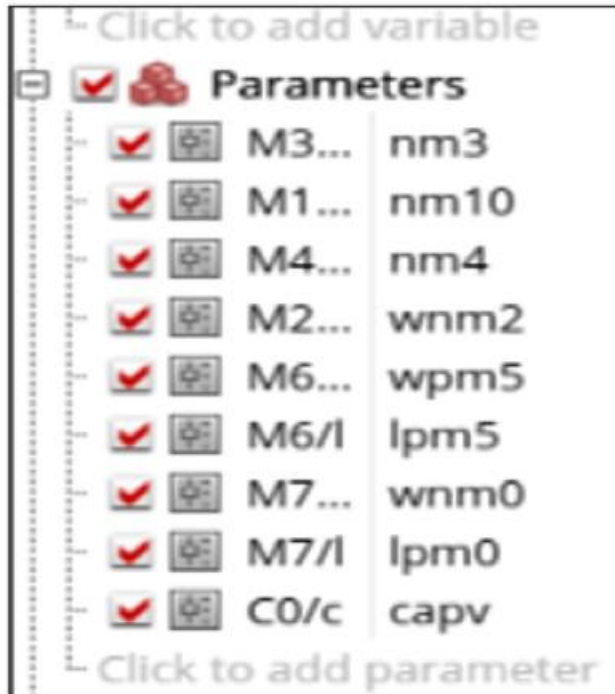
除了常见的design、test、corner外，重点设置

- ① Parameters、
- ② Global Variables、
- ③ Output
- ④ 选择Single Run



此处，必须选 single run, sweeps and corners

其他选择，优化器无法工作。



参数名要和电路图中的参数名对应

□ 可以手动写入.

□ 也可借用cadence 内部optimization 支持的工具写入.



(3.2) 添加、编辑variables

Global Variables		
☒	☒	nm3 3.084958616127238
☒	☒	nm... 13.16075831537826
☒	☒	nm4 5.877241929056585
☒	☒	wn... 8.97416483869768e-06
☒	☒	wp... 1.612683406299117e-05
☒	☒	lpm5 8.946240107792791e-06
☒	☒	wn... 3.899677099459485e-05
☒	☒	lpm0 7.859309281775453e-06
☒	☒	capv 6.417926634274258e-12
Click to add variable		

Data View	
Name	Value
Filter Filter	
Tests	
Global Variables	
☒	nm3 3.084958616127238
☒	nm... 13.16075831537826
☒	nm4 5.877241929056585
☒	wn... 8.97416483869768e-06
☒	wp... 1.612683406299117e-05
☒	lpm5 8.946240107792791e-06
☒	wn... 3.899677099459485e-05
☒	lpm0 7.859309281775453e-06
☒	capv 6.417926634274258e-12
Click to add variable	
Parameters	
☒	M3... nm3
☒	M1... nm10
☒	M4... nm4
☒	M2... wnm2

Add Variable

Edit Variable

Copy

Undo

Redo

Delete Variable

Backannotate to Test

Enable/Disable

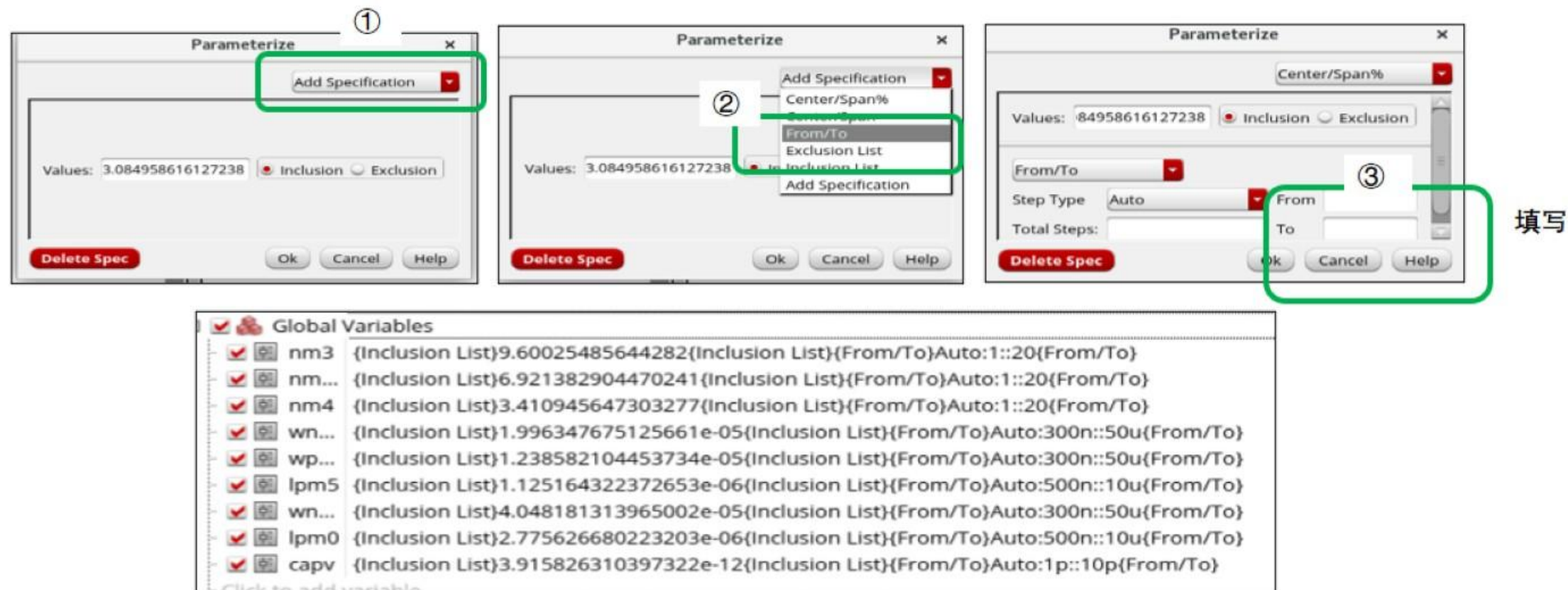
Group as Parametric Set

Add to Parametric Set

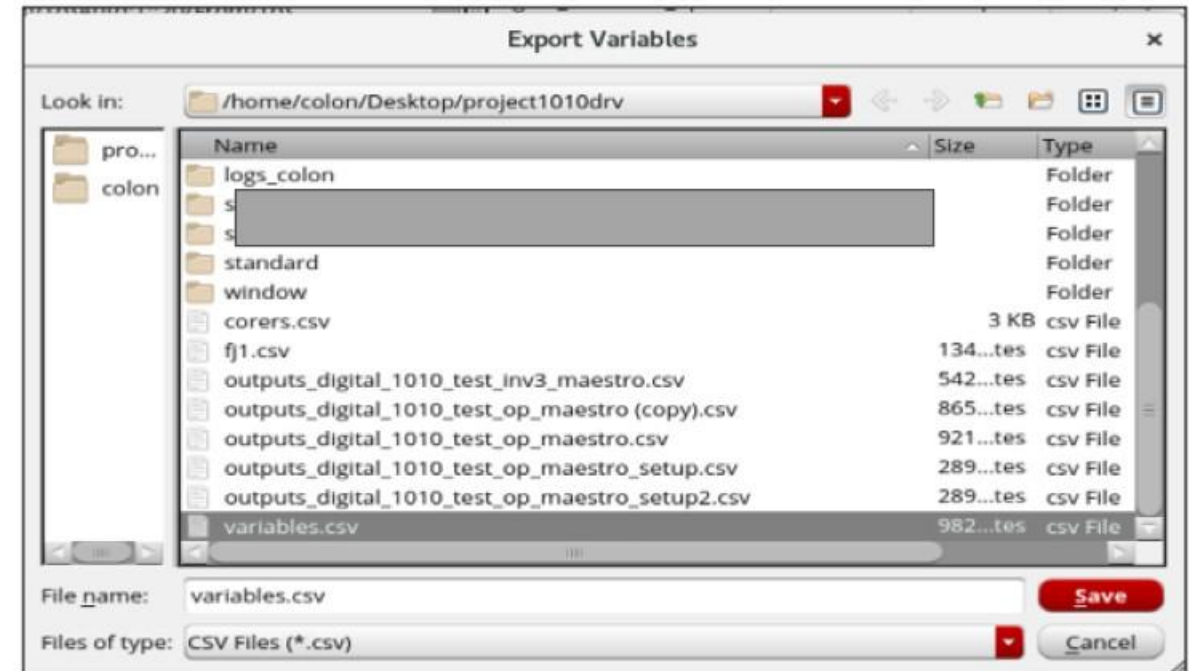
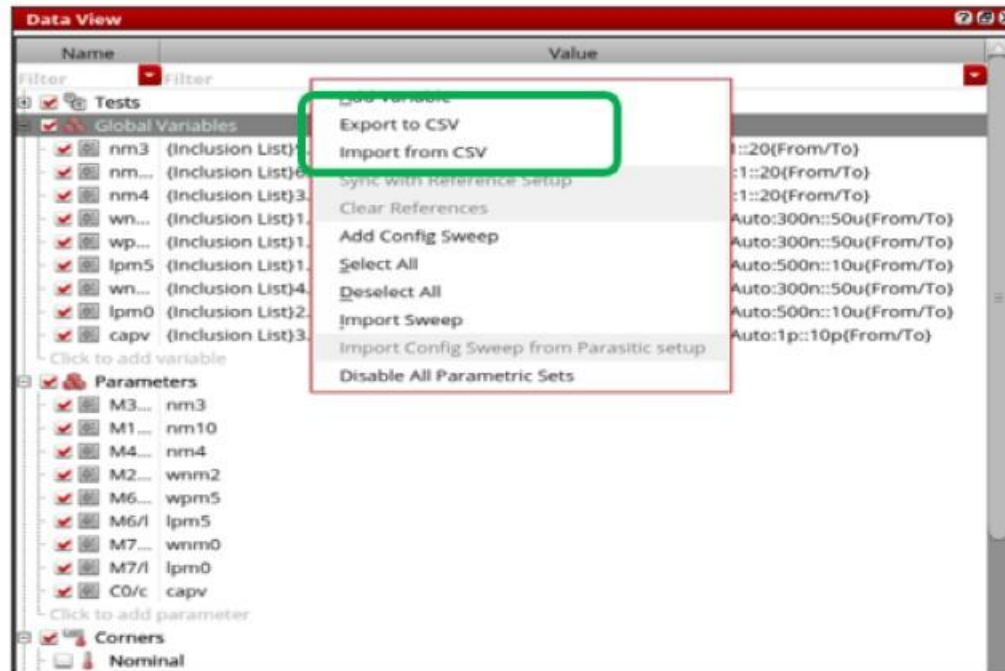
Import Sweep

Notes...

Delete Notes



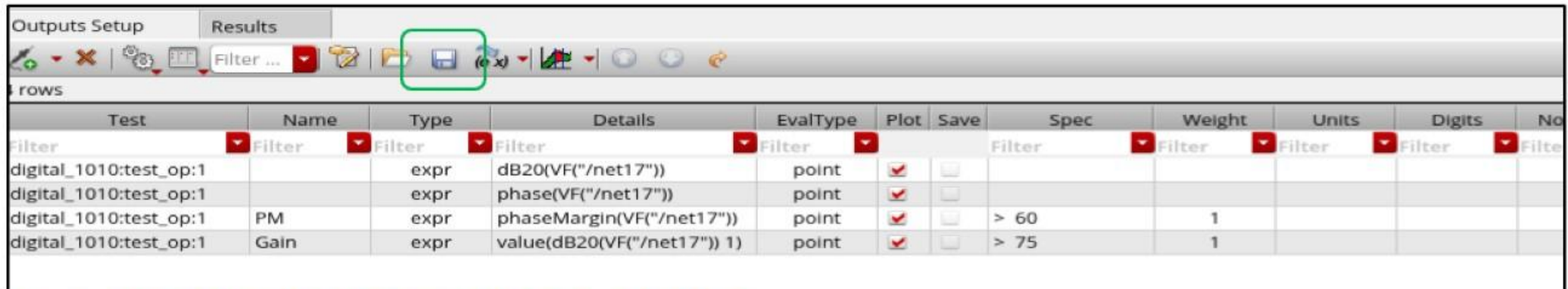
写完后，应为上图形式。



存储变量设置到variables.csv. 以备下次优化使用



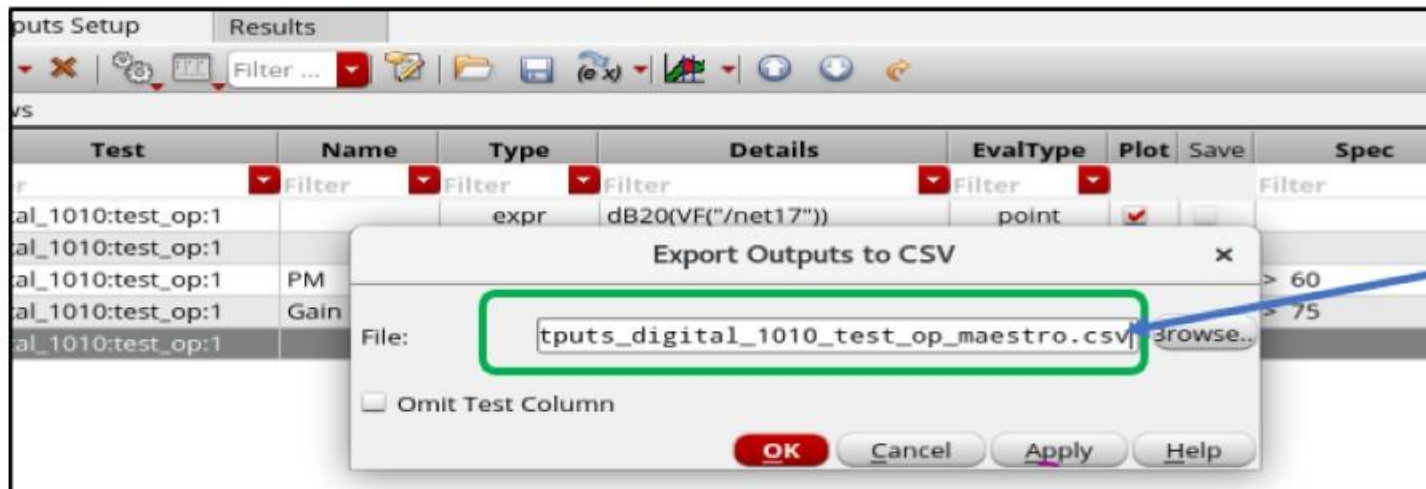
(3.3) Output 设置



Test	Name	Type	Details	EvalType	Plot	Save	Spec	Weight	Units	Digits	No
digital_1010:test_op:1		expr	dB20(VF("/net17"))	point	☒	☐					
digital_1010:test_op:1		expr	phase(VF("/net17"))	point	☒	☐					
digital_1010:test_op:1	PM	expr	phaseMargin(VF("/net17"))	point	☒	☐	> 60	1			
digital_1010:test_op:1	Gain	expr	value(dB20(VF("/net17")) 1)	point	☒	☐	> 75	1			

Output设置和通常设置并无重大的不同。但需注意

- ① 作为“优化目标”的项必须要填“Name”、“Weight”、“Weight”
- ② 无“Name”的expr类输出、和Signal类输出不被作为“优化目标”



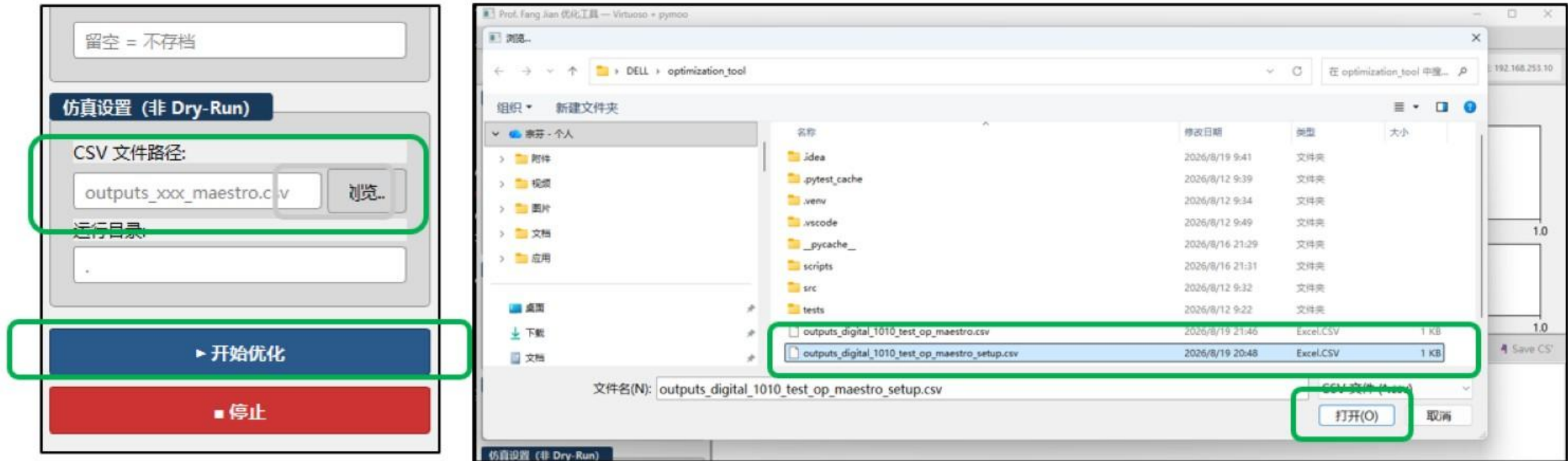
文件名必须修改

原名:

....._maestro.csv

改成:

....._maestro_setup.csv



Optimization 窗口

- ❑ 目前，脚本尚有缺陷，建议手动copy “..._maestro_setup.csv” 到windows 下面指定csv 文件路径及文件。
- ❑ 点击开始优化，优化器就开始工作了。



4. RUN

注：这是v0.1.0版本（暂未更新图片）

The screenshot displays the Prof. Fang Jian Optimization Tool interface, which is divided into two main sections: the Cadence window on the left and the Optimization window on the right.

Cadence Window: This section shows the Virtuoso ADE Assembler Editing interface. It includes a parameter table with columns for Parameter, it25, and it125. The table lists various parameters such as bc018_auto_v, mom_it, var_it, setup, soa_on, res_it, mim_it, ggmms_it, esdprp_it, dio_it, bjt_it, hve_it, and temperature. Below the parameter table, there is a table with columns for Test, Output, Spec, Weight, Pass/Fail, Min, Max, it25, and it125. The table shows results for three tests: digital_1010test_op:1, digital_1010test_op:1, and digital_1010test_op:1. The results are summarized in the following table:

Test	Output	Spec	Weight	Pass/Fail	Min	Max	it25	it125
digital_1010test_op:1	dB20VVF/me...							
digital_1010test_op:1	phaseVF/me...							
digital_1010test_op:1	PM	> 60	1	pass	89.38	97.22	89.38	97.22
digital_1010test_op:1	Gain	> 75	1	pass	84.06	85.93	84.06	85.93

Optimization Window: This section shows the optimization tool's configuration and results. It includes a menu bar (File, Optimization, View, Language, Help) and a status bar (Virtuoso: Connected, Disconnect, VNC: 未连接). The main area contains several panels:

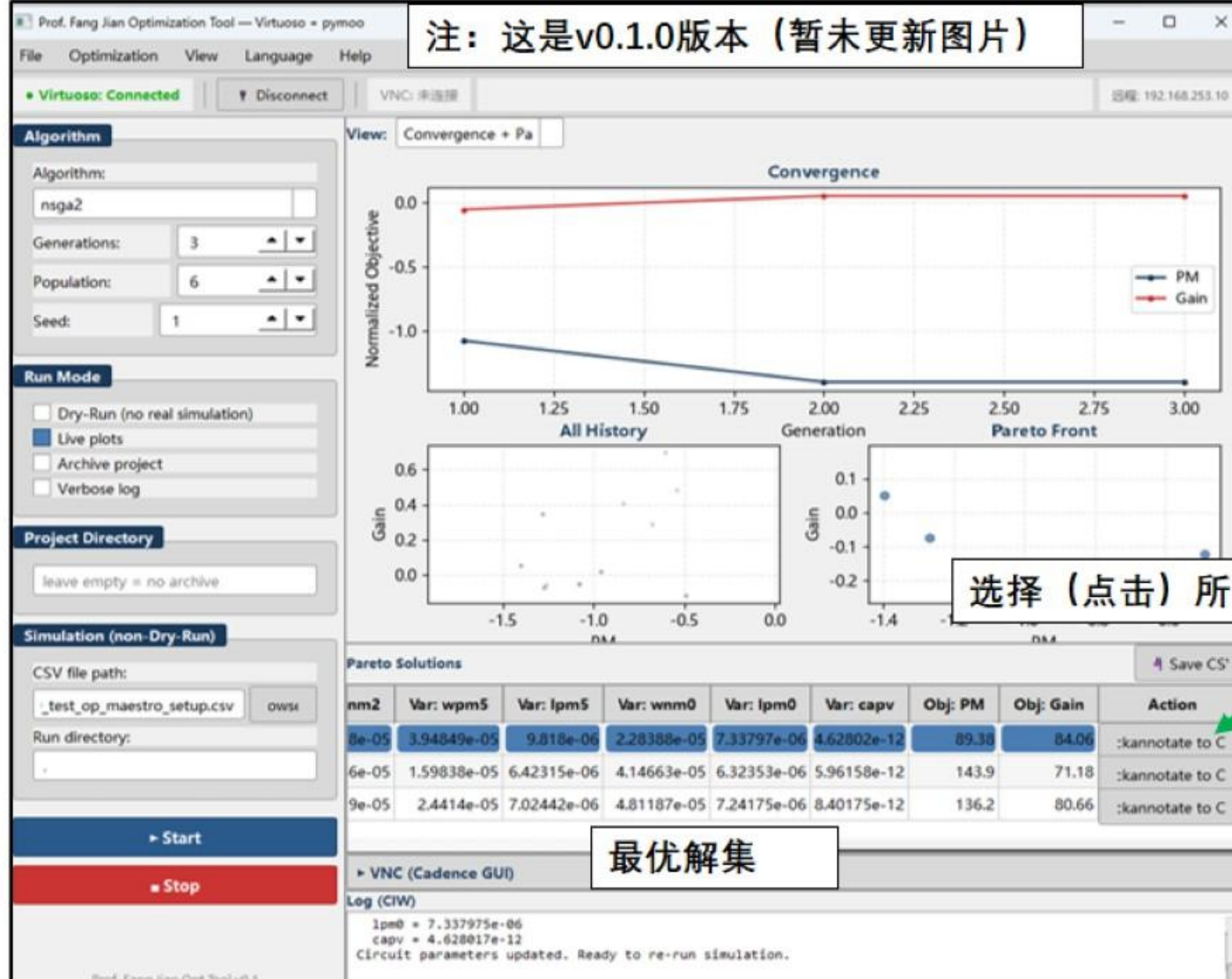
- Algorithm:** Algorithm: nsga2, Generations: 3, Population: 6, Seed: 1.
- Run Mode:** Dry-Run (no real simulation), Live plots, Archive project, Verbose log.
- Project Directory:** leave empty = no archive.
- Simulation (non-Dry-Run):** CSV file path: _test_op_maestro_setup.csv, Run directory: .
- Start/Stop Buttons:** Start (blue), Stop (red).
- View:** Convergence + Pa.
- Convergence Plot:** A line graph showing Normalized Objective vs. Generation. The PM (blue line) starts at -1.0 and decreases to -1.0. The Gain (red line) starts at 0.0 and increases to 0.0.
- All History Plot:** A scatter plot showing Gain vs. PM.
- Pareto Front Plot:** A scatter plot showing Gain vs. PM.
- Pareto Solutions Table:** A table showing the results of the Pareto front optimization. The table has columns for nm2, Var, wpm5, Var, lpm5, Var, wnm0, Var, lpm0, Var, capv, Obj: PM, Obj: Gain, and Action. The table shows three solutions: 8e-05, 6e-05, and 9e-05.
- Log (CIW):** A text area showing the optimization log, including the parameters lpm0 = 7.337975e-06 and capv = 4.628017e-12.

Cadence 窗口

Optimization 窗口



5. Result & Callback



Name Filter Filter

Tests

Global Variables

nm3	1.752264656619854
nm...	11.04319600194041
nm4	9.727381774822671
wn...	3.398774083748817e-06
wp...	3.217401000622693e-05
lpm5	8.600011965566239e-06
wn...	2.976916859978291e-05
lpm0	2.97092575350362e-06
capv	8.558933689282679e-12

Click to add variable

Parameters

回调至电路图

被刷新

选择（点击）所需的最优解，回传到cadence。